



EVALUATING MULTIMODAL MULTIPLAYER GAMES FOR COGNITIVE & SOCIAL SKILLS DEVELOPMENT IN DYSLEXIC CHILDREN

25-26J-421

OUR TEAM



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Ethical Clearance



SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

16th Floor, BoC Merchant Tower, No. 28, St. Michael's Road, Colombo 03

Date : 1 March 2026

Your Ref :

My Ref : FoCERC25-0035

From: SLIIT - Ethics Review Committee

Mr. H.M.K.M. Bandara (IT22562142)
Faculty of Computing, SLIIT

Dear Mr. H.M.K.M. Bandara,

Status of Your Ethics Review Application

This is with reference to your request on Ethical Clearance of students' research projects under the name of "*Evaluating Multimodal Multiplayer Games for Cognitive and Social Skills Development in Dyslexic Children*". The Ethics Review Committee (ERC) has reviewed the application, and the proposal is considered to meet the requirement of the ethics related concerns accepted by the Committee members. Accordingly, the ERC has given the following decision for the below stated project.

Project Title	<i>Evaluating Multimodal Multiplayer Games for Cognitive and Social Skills Development in Dyslexic Children</i>
Supervisors	Mr. Didula Chamara, Faculty of Computing, SLIIT Mr. Nushkan Nisme, Faculty of Computing, SLIIT Dr. Kasun Karunanayaka, University of Colombo School of Computing
Co-Investigators	Bandara H.M.K.M. (IT22562142) Samarasekara D.P. (IT22599704) Ranasinghe S.P. (IT22597656) Gunasekara H.T.P.D.N.P. (IT22610034)
ERC Decision	Approved
Date of Decision	17 February 2026

Kindly note that the Committee wanted you to ensure that the conduct of the research project is strictly in accordance with the approved proposal. Please proceed with the research as per the approved protocol. Any modifications to the research must be reported to the Ethics Review Committee (ERC) for further approval. We appreciate your commitment to conducting research in accordance with ethical standards. Should you have any questions or require further assistance, please do not hesitate to contact us at foc.erc@slit.lk.

Thank you for your cooperation.
Sincerely,

Prof. Dasuni Nawinna
Chair/ FoC - Ethics Review Committee, Faculty of Computing - SLIIT

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The Word Garden: Design, Development, and Expert Evaluation of a Multimodal Multiplayer Literacy Game for Children with Dyslexia

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Abstract—Children with dyslexia face not only literacy challenges but also heightened anxiety and diminished motivation from repeated failure. Existing digital interventions prioritize phonological drills while under-addressing emotional safety and social support, operating predominantly as single-user applications. This paper presents The Word Garden, a cooperative multiplayer educational game employing dual-device architecture to integrate emotionally safe social interaction with literacy skill development for children aged 8-12 with dyslexia. The system separates collaborative PC-based exploration from private smartphone-based multisensory literacy tasks, enabling children to practice challenging exercises without peer visibility while maintaining social connection through voice communication. The design emphasizes cooperative mechanics without competitive elements, low-visibility interaction, and non-punitive feedback to reduce performance anxiety. Expert validation by consultant child and adolescent psychiatrists confirmed alignment between design features and therapeutic principles for literacy intervention. Technical evaluation demonstrated system feasibility with stable multiplayer synchronization, 98.3% event delivery accuracy, and zero crashes across 48 hours of testing with 2-5 concurrent players. This work constitutes Phase 1 of a longitudinal study; child user evaluation and empirical learning outcomes will be addressed in Phase 2. This formative evaluation establishes technical and design foundations for investigating whether cooperative multiplayer environments can effectively support both literacy development and social-emotional well-being in children with dyslexia.

Index Terms—Dyslexia intervention, Multisensory learning, Educational game design, Smartphone-based learning, AI-assisted literacy learning

I. INTRODUCTION

Dyslexia, a neuro-developmental disorder affecting 5-10% of children globally [1], is characterized by persistent deficits in word recognition, decoding, and spelling despite adequate

intelligence and instruction [2]. Beyond academic challenges, affected children experience heightened anxiety, diminished self-esteem, and reduced reading motivation from repeated failure [3].

Traditional phonics-based interventions demonstrate short-term gains but fail to sustain engagement due to their monotonous, high-pressure nature, often exacerbating fear of errors and low self-competence [1], [3]. Recent advances in immersive technologies have shown potential to modernize education through interactive, risk-free learning environments [4]. Game-based approaches offer particular promise by reducing failure anxiety through playful mechanics and fostering intrinsic motivation [5].

However, existing dyslexia interventions under-address motivation and emotional safety, prioritizing cognitive drills over affective factors [3]. Current digital tools are predominantly single-user applications lacking social support, creating isolated experiences [1]. While game-based learning shows promise for literacy [6], [7], multiplayer literacy games targeting dyslexia remain rare [6]. Existing approaches involve individual play with incidental social interaction rather than networked cooperative gameplay with intentional collaborative mechanics addressing both literacy and social-emotional needs [8].

This paper introduces "The Word Garden," a cooperative multiplayer educational game employing dual-device architecture for dyslexic children, featuring PC-based collaborative exploration and smartphone-based private literacy tasks. This architecture enables cognitive compartmentalization while maintaining emotional safety. Children engage in literacy exercises on personal devices without peer visibility, reducing performance anxiety while preserving cooperative gameplay benefits. The system incorporates: (1) dual-device architecture com-

Our paper was published at the 9th IEEE International Research Conference on Smart Computing and Systems Engineering (SCSE) held at the University of Kelaniya.



WHAT IS DYSLEXIA?

A neurodevelopmental learning difficulty affecting accurate word recognition, letter decoding, spelling, and phonological processing. Impacts ~5–10% of children worldwide. Also causes **reduced self-confidence, anxiety, fear of failure, and social withdrawal.**

BECAUSE OF DYSLEXIA?

Children struggle to read and decode letters, have poor working memory, and find it hard to keep up in fast-paced digital environments. Beyond academics, they face social communication difficulties such as **fear of making mistakes in front of peers, withdrawal from group learning, and low self-belief** creating cycles of **isolation and disengagement.**

RESEARCH GAP

No game-based solution exists that targets these **social dimensions**. On the technology side, no system combines multiplayer + multimodal + dual-device (mobile + PC) interaction for dyslexic children. Tools rely heavily on post-error feedback, lack voice-enabled peer interaction, and ignore social confidence entirely.

RESEARCH PROBLEM

Current digital tools for dyslexia are mostly single-player, focusing only on decoding accuracy while **ignoring social and emotional needs**. They rely on post-error feedback that reinforces failure. No existing system combines multiplayer collaboration, cross-device interaction (mobile + PC), and voice-enabled peer communication in a literacy game designed specifically for dyslexic children.

SOLUTION

A real-time multiplayer literacy game - PC environment (Unreal Engine) + mobile companion app (Flutter) — with voice chat, anticipatory feedback, and non-competitive rewards **to build literacy, social confidence, and reduce fear of failure.**



MAIN OBJECTIVES

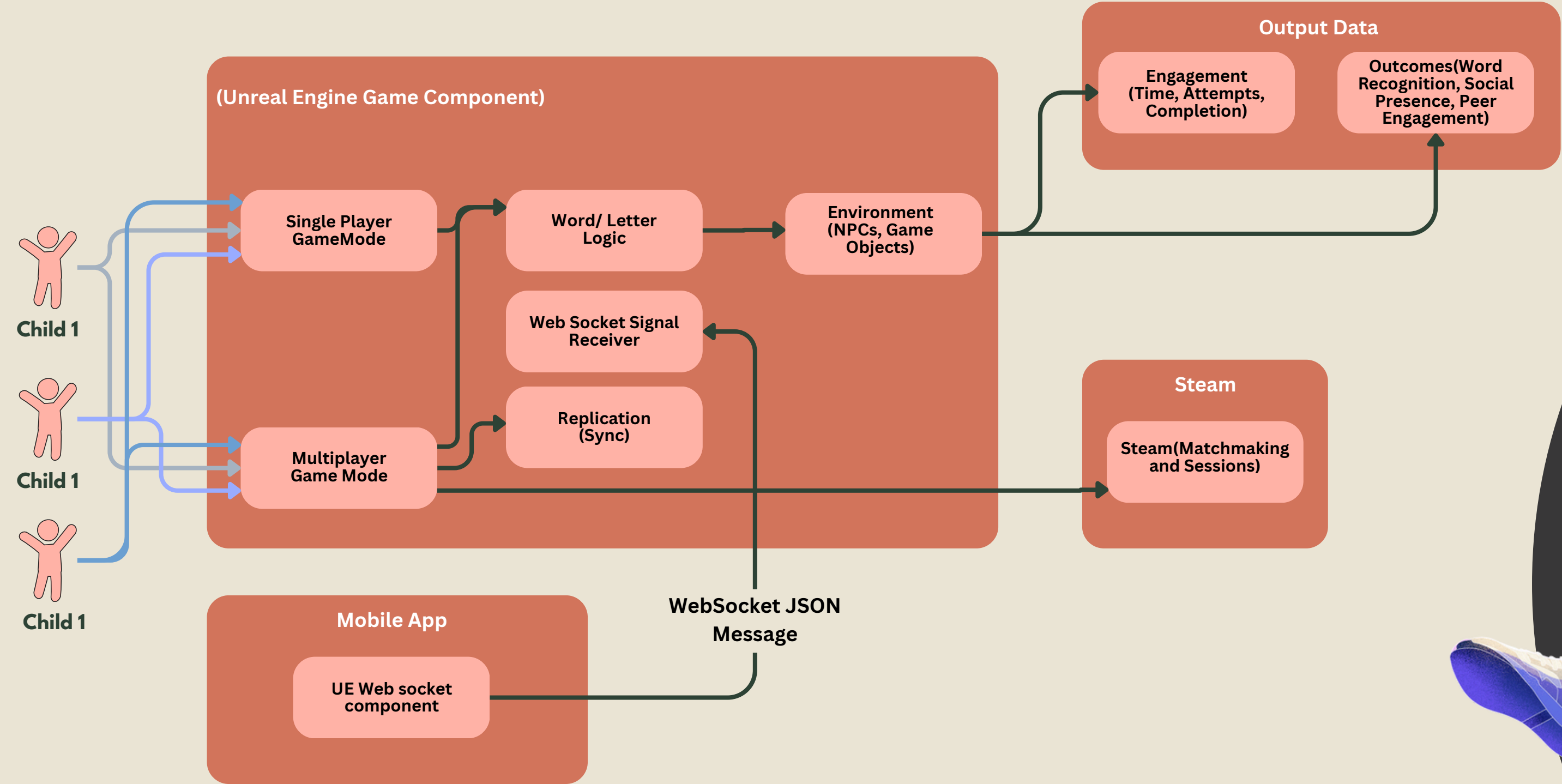
- **Objective 1** - Design a **mobile app** for structured letter tracing & selection to support **motor memory** and **personalized learning**.
- **Objective 2** - Develop a real-time **multiplayer virtual environment** for **collaborative literacy tasks** and **peer interaction**.
- **Objective 3** - Evaluate the impact of **multimodal multiplayer** gameplay on **attention, engagement, and learning** vs. **single-player tools**.
- **Objective 4** - Incorporate **anticipatory feedback** to guide learners before errors occur, **reducing frustration** and **improving accuracy**.
- **Objective 5** - Integrate **voice communication** and **non-competitive rewards** to **boost motivation, social confidence, and emotional comfort**.



SYSTEM ARCHITECTURE

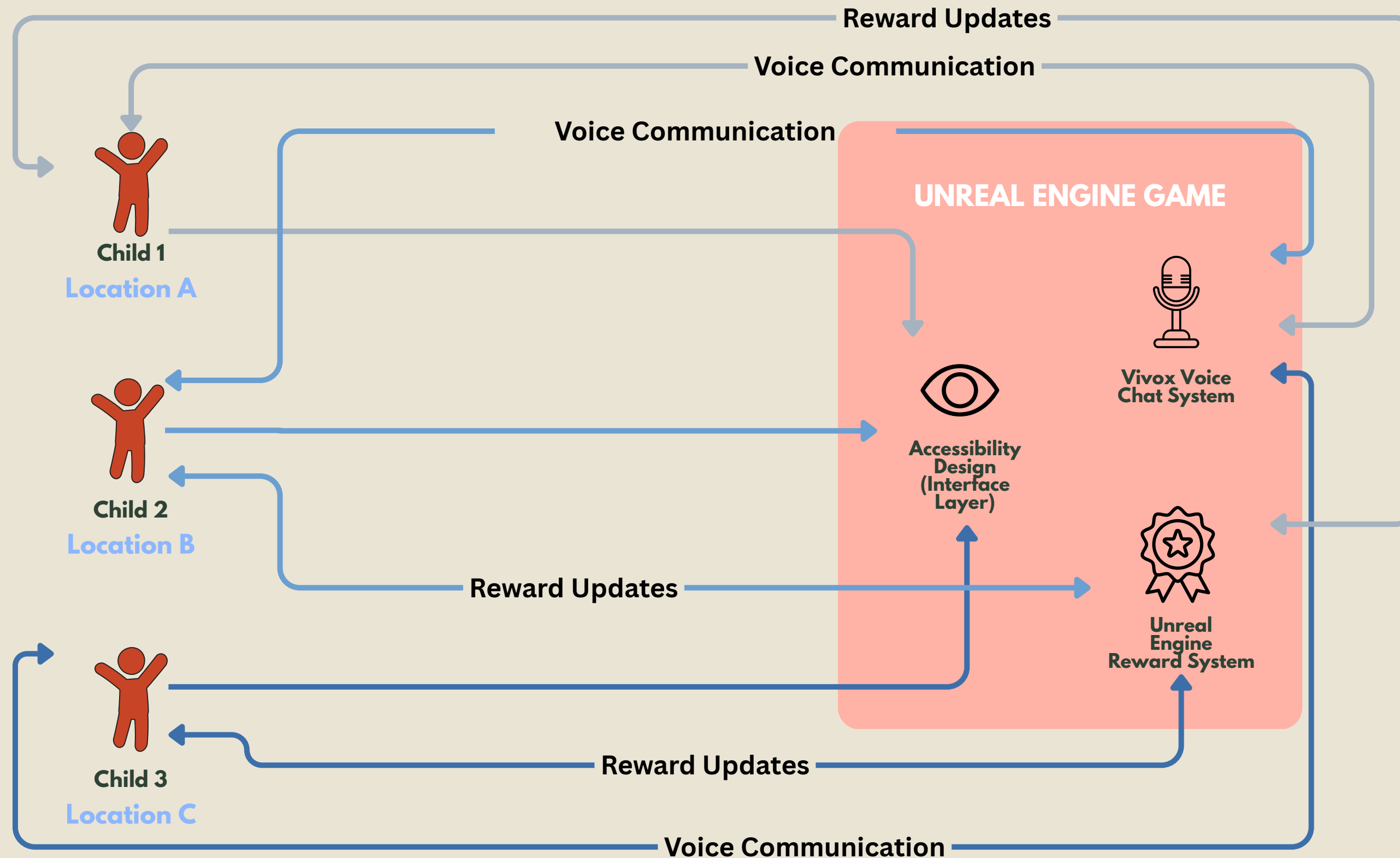
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MULTIMODAL GAMEPLAY MODULE



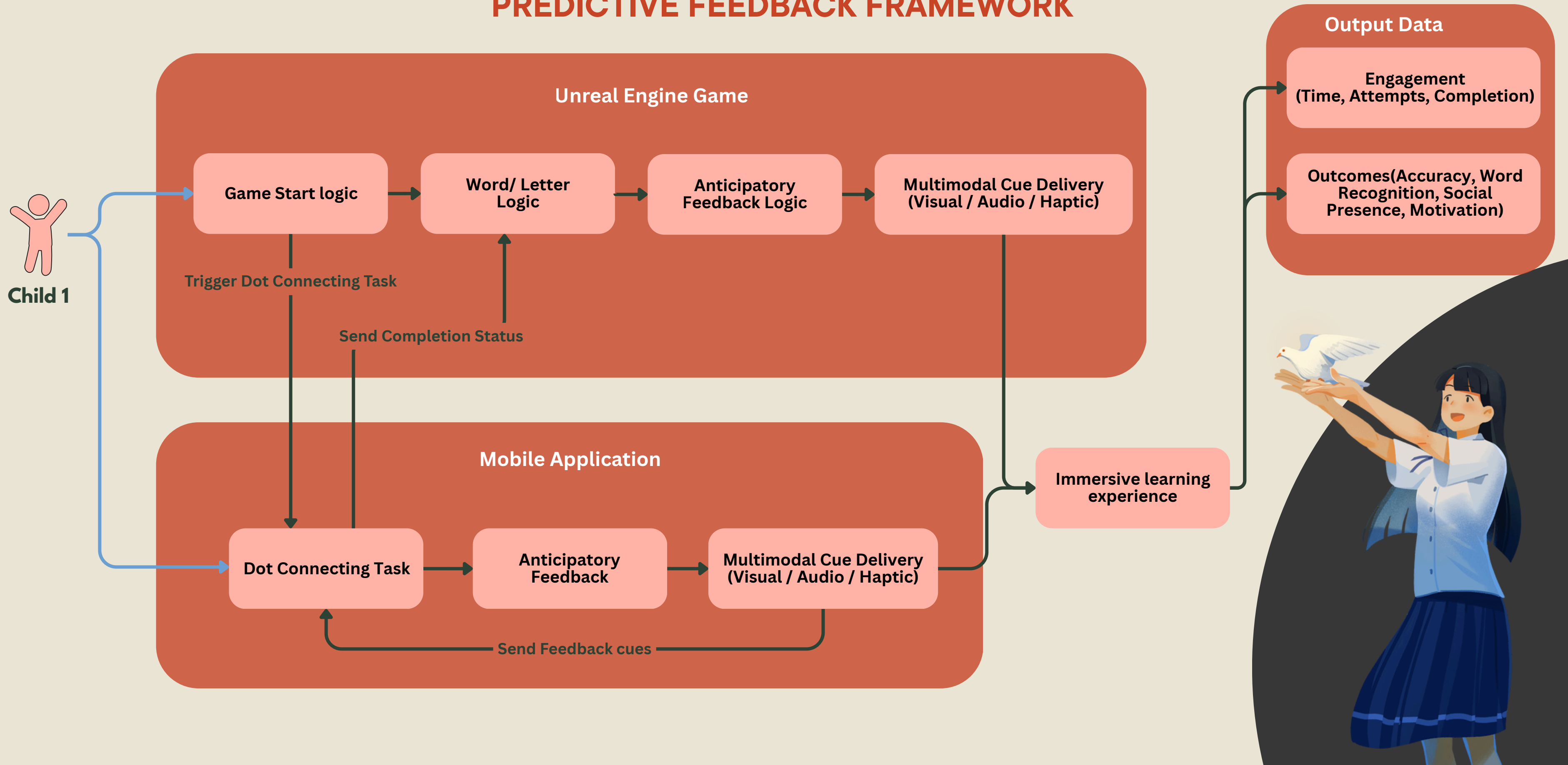
SAMARASEKARA D.P. - IT22599704

SOCIAL ENGAGEMENT SYSTEM



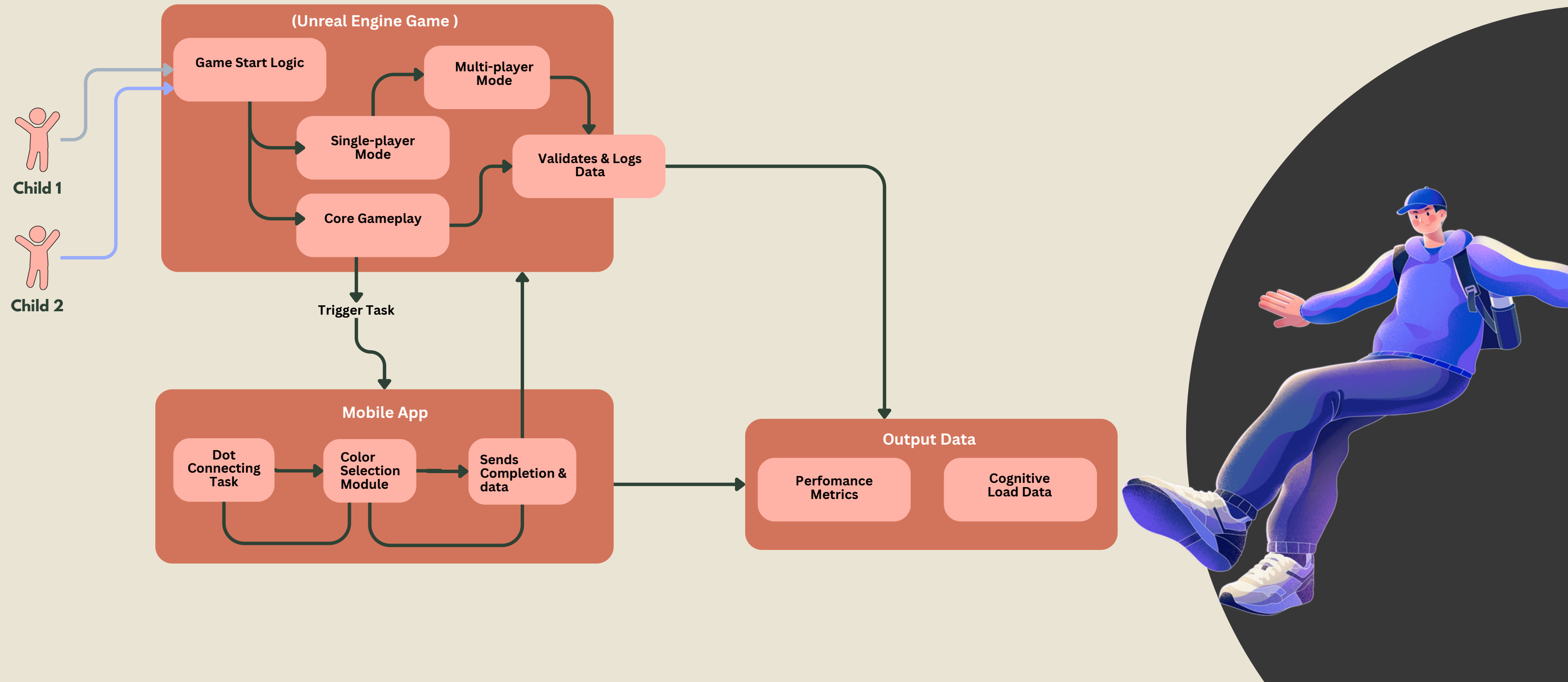
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PREDICTIVE FEEDBACK FRAMEWORK



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COGNITIVE INTERACTION ENGINE



COMMERCIALIZATION

WHO WILL BENEFIT?

- Children with dyslexia & literacy challenges.
- Schools (primary & secondary)
- Special education & therapy centers
- Parents (home-based learning)
- Educational NGOs & government programs
- EdTech platforms expanding into accessibility



VALUE PROPOSITION

- Combines literacy learning + multiplayer motivation.
- Inclusive & engaging: dyslexia-friendly fonts, multimodal prompts, adaptive paths.
- Reduces isolation, boosts peer collaboration & motivation.
- Future-ready: cross-platform, VR/AR, localization.

INTELLECTUAL PROPERTY RIGHTS (IPR)

- Proprietary framework design (multiplayer literacy model).
- Trademarked game title & branding.
- Copyright-protected content & assets.
- Potential for patents (unique multiplayer literacy learning mechanism).

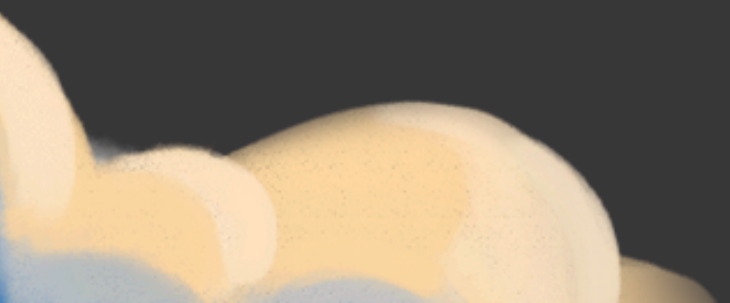
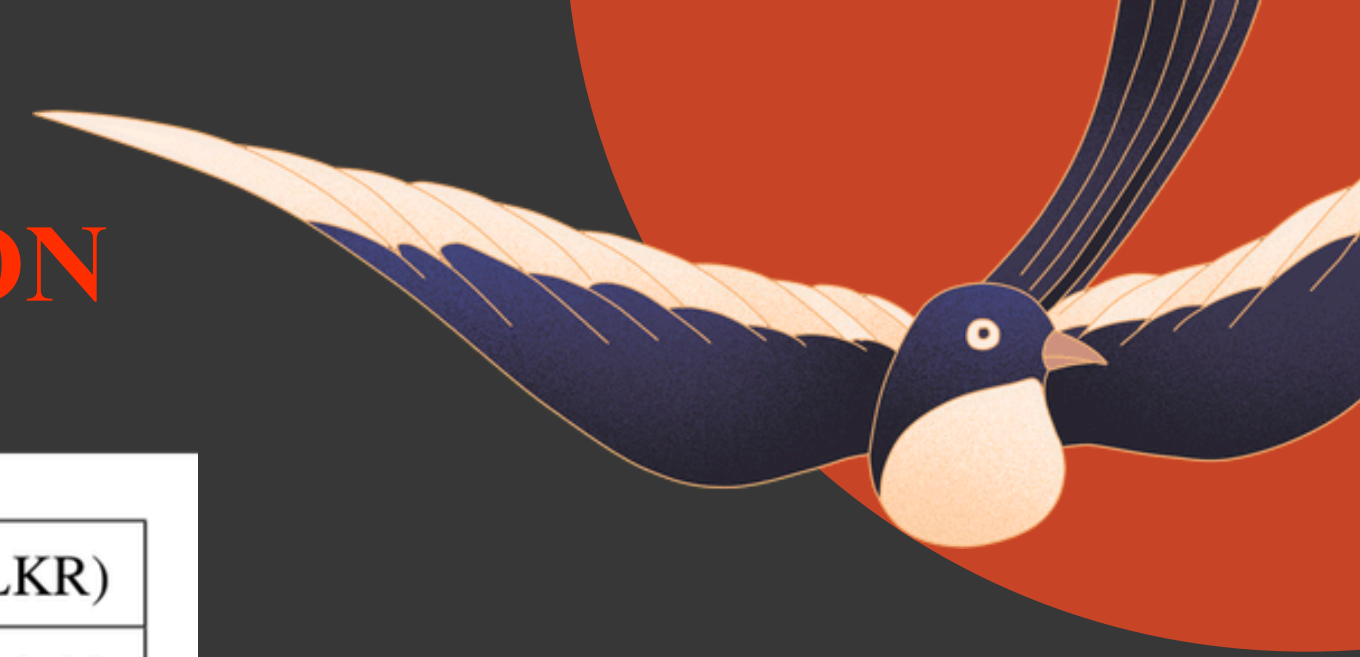
REVENUE MODEL

- **Direct Sales (Steam)**
(Steam, world-wide, low-cost entry)
- **Institutional Licensing**
(schools/therapy centers bulk adoption)
- **Grants & Partnerships**
(NGOs, government)
- **Premium Features (DLC)**
(analytics, adaptive learning, multiplayer classrooms, new maps)



BUDGET JUSTIFICATION

Requirement	Cost Approx. (LKR)
Development & Testing Resources	20,000.00
Participant Incentives	10,000.00
Printing & Research Materials	5,000.00
Internet & Server Costs	5,000.00
Travelling Cost	5,000.00
Ethical Approval Cost	2,000.00
Publishing Cost	15,000.00
Contingency / Miscellaneous	5,000.00
Consultant Cost	10,000.00
Total Cost	77,000.00



DEMONSTRATION



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EVALUATING THE IMPACT OF MULTIMODAL MULTIPLAYER VERSUS TRADITIONAL SINGLE-PLAYER GAMES ON DYSLEXIC CHILDREN'S LEARNING

OBJECTIVES AND HOW THEY ARE ADDRESSED

Objective	How the System Overcomes It
Evaluate the impact of multiplayer vs. single-player literacy gameplay on learning and social outcomes	A controlled prototype with both modes built in UE5 allows direct comparison of engagement, motivation, and task performance
Investigate multimodal interaction (drawing, visual recognition, peer collaboration) in enhancing literacy	Drawing-based word-building via mobile app sends input through WebSocket to UE5, combining motor, visual, and social engagement
Examine whether multiplayer environments address social challenges like low confidence and peer withdrawal	Real-time collaboration and competition mechanics encourage positive peer interaction, reducing isolation through shared gameplay
Provide empirical evidence on multiplayer vs. single-player effectiveness in attention, error reduction, and retention	Data logging and telemetry track accuracy, retries, and session performance across multiple sessions for scientific comparison
Generate practical design insights for next-gen inclusive educational games	Findings inform accessibility-first design guidelines combining dyslexia-friendly UI, multimodal input, and multiplayer collaboration

COMPARISON OF PREVIOUS RESEARCHES

Application Features	Research A (GraphoLearn – Single-player Literacy Game) [6]	Research B (iLearnRW – Adaptive Reading Support) [7]	Research C (Words Matter – Collaborative Dyslexia Game) [4]	Proposed System (Multimodal Multiplayer Literacy Framework)
Real-time multiplayer interaction	✗	✗	✓	✓
Peer collaboration and social presence	✗	✗	✓	✓
Immersive environment	✗	✗	✗	✓
Adaptive literacy feedback for dyslexic learners	✓	✓	✗	✓
Multimodal support (visual + auditory cues)	✓	✓	✗	✓
Dyslexia-friendly accessibility (fonts, contrast, narration)	✓	✓	✗	✓
Cross-platform / mobile accessibility	✗	✓	✗	✓
Learning analytics & progress tracking	✗	✓	✗	✓

COMPARISON OF PREVIOUS APPLICATIONS

Requirement	Strengths	Limitations	Identified Gap
Phonics-based interventions [6], [7]	Improves decoding, early literacy	Single-user, limited engagement	Lack of social/peer interaction
Dyslexia definitions & theories [8]-[12]	Strong theoretical basis (phonological, cerebellar, grain size)	Limited translation into interactive digital tools	Need for applied interventions
Gamified literacy tools [1]	Boosts motivation and reading outcomes	Short-term focus, single-player	Missing collaborative dynamics
Digital gamebased training [2], [3]	Cognitive and literacy gains across NDDs	Not dyslexiaspecific in multiplayer contexts	Lack of targeted multiplayer trials
GraphoLearn [6]	Adaptive, crosslinguistic assessment + intervention	Designed for individual users	No real-time peer collaboration
Multiplayer game (Words Matter) [4]	Promotes peer learning and social confidence	Prototype stage, limited empirical validation	Further evaluation in controlled trials



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**VOICE-ENABLED LEARNING & REWARDS FOR DYSLEXIC CHILDREN:
ENHANCING COMMUNICATION, MOTIVATION & SELF-REGULATION**

OBJECTIVES AND HOW THEY ARE ADDRESSED

Objective	How the System Overcomes It
Develop a voice-enabled multiplayer FPS game with real-time chat and reward system	Built on Unreal Engine with Vivox plugin for low-latency voice chat and synchronized rewards across players
Facilitate dynamic peer collaboration through voice communication	Real-time network sync enables verbal strategy discussions, building social confidence and reducing isolation
Support cognitive skills - attention, working memory, meta-cognitive awareness	Letter-shooting mechanic targets sequencing and attention indirectly, without phonological training
Enhance motivation and self-regulation through adaptive rewards	Backend tracks accuracy and task completion, delivering instant rewards while adaptive difficulty prevents overwhelm
Incorporate dyslexia-friendly accessibility features	High-contrast visuals, OpenDyslexic fonts, noise-reduced audio, and customizable UI based on individual profiles

COMPARISON OF PREVIOUS RESEARCH

Application Features	Research A (Action Video Games e.g., FruitNinja) [3]	Research B (Adaptive Serious Games) [4]	Research C (Verbal Collaboration Approaches) [7]	Proposed System (Multimodal Multiplayer Literacy Framework)
Real-time multi player support	✗	✗	✓	✓
Real-time voice communication	✗	✗	✓	✓
Multimodal cues (visual + audio)	✓	✓	✗	✓
Social interaction & peer collaboration	✗	✗	✓	✓
Gamification (positive rewards, badges)	✓	✓	✗	✓
Accessibility (fonts, contrast, audio support)	✗	✓	✗	✓

COMPARISON OF PREVIOUS APPLICATIONS

Requirement	Strengths	Limitations	Identified Gap
Voice-enabled multiplayer FP game [7]	Builds peer talk, teamwork, and confidence.	Prototype stage, not fully tested.	Lacks large-scale validation.
Gamified reward system (badges, stars, tokens) [4]	Boosts motivation, persistence, and self-control.	May cause over-reliance on rewards.	Few long-term motivation studies.
Dyslexia-friendly features (fonts, contrast, audio) [17]	Increases usability and reduces strain.	May be underused without guidance.	No standard game design guidelines.
Social-motivational dynamics [6]	Encourages peer learning and cooperation.	Multiplayer may face tech issues.	Few real-time collaborative studies.



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ANTICIPATORY MULTIMODAL FEEDBACK SYSTEM IN A MULTIPLAYER
DYSLEXIC GAME

OBJECTIVES AND HOW THEY ARE ADDRESSED

Objective	How the System Overcomes It
Design anticipatory multimodal cues (visual, audio, haptic) triggered before errors occur	Mobile app detects drawing deviation in real time → fires glowing outlines, phoneme tones & vibrations
Compare accuracy & error rates between anticipatory (Group A) vs post-error feedback (Group B)	Two switchable feedback modes; 30 dyslexic children randomly assigned; errors auto-logged in-game
Measure motivation via gameplay time, attempts & self-reported enjoyment	Multiplayer + voice chat + leaderboards drive engagement; logging captures session stats & surveys
Evaluate learning retention at post-task, 1 day & 1 week intervals	Three-phase testing schedule embedded in the experiment; data stored for longitudinal comparison
Assess cognitive load using NASA-TLX (child-adapted) to check if anticipatory cues reduce mental effort	Post-session NASA-TLX surveys integrated in-game; analytics dashboard runs ANOVA comparisons

COMPARISON OF PREVIOUS RESEARCHES

Application Features	Post-Error Feedback	Anticipatory Feedback	Multimodal Cues (Visual + Audio + Haptic)	Multiplayer Literacy game
Research A Shaywitz (2003) [2]	✓	✗	✗	✗
Research B Kroesbergen & Van Luit (2003) [3]	✓	✗	✗	✗
Research C Ma et al. (2021) [5]	✓	✗	✓	✗
Research D Conati & Kardan (2013) [8]	✓	✓	✗	✗
Research E Vahabzadeh et al. (2019) [9]	✓	✗	✓	✓
Proposed System	✓	✓	✓	✓

COMPARISON OF PREVIOUS APPLICATIONS

Requirement	Strengths	Limitations	Identified Gap
Vahabzadeh et al. (2019) – VR-Based Learning Games [9]	Boosts engagement & retention	Limited dyslexia focus, post-error only	Missing anticipatory feedback in multiplayer dyslexia learning
Hamari & Koivisto (2015) – Game-Based Learning Review [10]	Improves motivation & collaboration	Few anticipatory feedback systems	No real-time anticipatory cues in multiplayer dyslexia games
Conati & Kardan (2013) – Multimodal AI Agents [8]	Early success in error prevention	Limited dyslexia application	No real-time multiplayer anticipatory feedback for dyslexia
Shaywitz (2003) – Dyslexia Training Apps [2]	Widely used, effective basics	Mostly single-player, no anticipatory feedback	No cross-platform multiplayer anticipatory feedback



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PERSONALIZED DRAWING AND COLOR SELECTION
WITH MULTISENSORY FEEDBACK IN A MULTIPLAYER DYSLEXIC GAME

OBJECTIVES AND HOW THEY ARE ADDRESSED

Objective	How the System Overcomes It
Design a customizable drawing interface for dyslexic children	Mobile app allows selection of colors, stroke width, and dot spacing tailored to individual sensory preferences
Implement real-time adaptive multisensory feedback	Visual (glowing strokes), auditory (TTS prompts), and haptic (vibration) cues guide letter formation and prevent errors
Integrate mobile drawing module with Unreal Engine multiplayer game	WebSocket/REST API syncs tracing completion signals to trigger in-game events like letter chests and magical objects
Assess impact on cognitive load, engagement, and literacy outcomes	Controlled study with 30 dyslexic children (ages 7–12) using NASA-TLX, SUS questionnaires, and tracing accuracy logs
Refine system based on user performance and feedback	Iterative usability testing with dyslexic learners informs adaptive calibration of feedback intensity and UI improvements

COMPARISON OF PREVIOUS RESEARCHES

Application Features	Research A (Ekawati et al. 2024) [1]	Research B (Blomert et al. 2014) [6]	Research C (Klinkenberg et al. 2011) [7]	Proposed System (Multimodal Multiplayer Literacy Framework)
Real-time multiplayer interaction	✗	✓	✗	✓
Balances personalization & fairness	✗	✗	✗	✓
Dynamic difficulty adjustment	✗	✗	✓	✓
Adaptive & personalized content	✗	✗	✓	✓
Measures cognitive load & engagement	✗	✗	✗	✓
Dyslexia-friendly accessibility (fonts, contrast, narration)	✓	✗	✗	✓
Real-time collaboration	✗	✓	✗	✓

COMPARISON OF PREVIOUS APPLICATIONS

Requirement	Strengths	Limitations	Identified Gap
Tasks matched to literacy levels (Ekawati et al. 2024) [1]	Effective in single-player skill building	Largely absent in multiplayer games	Multiplayer games apply "one-size-fits-all" approach
Dynamic complexity adjustment (Thesai.org 2022) [2]	Maintains challenge, reduces frustration	Rarely integrated in real-time multiplayer	Lack of dynamic difficulty scaling in collaborative multiplayer
Fairness & collaboration (PMC 2022) [4]	Promotes social motivation through shared goals	Personalization lacking leads to unfair player experiences	No mechanisms balancing challenge and team fairness
Supportive, non-stigmatizing feedback (Dyslexia Blog 2018) [3]	Tailored private feedback in single-player	Public feedback in multiplayer exposes skill disparities	Absence of private, supportive feedback in multiplayer
Social connectivity (Abdous 2023 [5])	Multiplayer promotes interaction	Personalized learning paths missing	Need for individualized learning paths within social frameworks

TEST RESULTS

TECHNICAL TESTING OUTCOMES

Metric	Result
Frame Rate (Discrete GPU)	70–80 FPS
Frame Rate (Integrated GPU)	30–35 FPS
Network Latency (2 players)	60 ms
Network Latency (5 players)	150 ms
Bandwidth per User (2 players)	140 kbps
Bandwidth per User (5 players)	450 kbps
PC–Smartphone Event Latency (Firebase)	160–190 ms
Task Delivery Accuracy	>98%
Session Stability (48 hrs, 2–5 players)	0 crashes

EXPERT VALIDATION OUTCOMES (5-POINT SCALE)

Design Dimension	Rating	Key Feedback
Emotional Safety Design	4.5 / 5	Low-visibility interaction & non-punitive feedback effectively reduce performance anxiety
Visual Design	4.5 / 5	Dyslexia-friendly colors, fonts & 3D letters viewed favorably
Cooperative Mechanics	5.0 / 5	Shared goals without competition align well with therapeutic principles
Dual-Device Architecture	4.5 / 5	Effectively balances collaborative engagement with private practice
Multisensory Integration	5.0 / 5	Visual, auditory & tactile modalities align with evidence-based dyslexia approaches
Implementation Feasibility	4.0 / 5	Requires facilitator training & session duration management



QUESTION & ANSWER



THANK YOU

Research Group

25-26J-421

